

Advanced News Recommender System

1. Abstract

This project presents an Advanced News Recommender System that utilizes natural language processing and machine learning techniques to recommend relevant news articles based on content similarity and sentiment analysis. The system combines TF-IDF vectorization, cosine similarity, sentiment polarity detection, and keyword extraction to create a personalized and context-aware recommendation framework. The model demonstrates how modern text analytics can be applied to improve information retrieval and user engagement in the news domain.

2. Introduction

In the era of digital information, the amount of content produced every second is overwhelming. Readers often struggle to identify the most relevant and interesting news articles among vast datasets. Recommender systems have emerged as an essential solution to this challenge. They help users navigate large volumes of data by automatically suggesting relevant items based on user behavior or content similarity. This project focuses on developing a content-based news recommender system enhanced with sentiment filtering, which ensures that users are recommended not only similar but also emotionally aligned news articles.

3. Objectives

The main objectives of the Advanced News Recommender System are as follows:

- To design a content-based recommendation model for news articles.
- To preprocess textual data effectively using NLP techniques.
- To apply TF-IDF vectorization and cosine similarity to compute article similarity.
- To analyze the sentiment polarity of each article using TextBlob.
- To extract significant keywords for better content understanding.
- To combine sentiment filtering with similarity-based recommendations to enhance result quality.

4. Literature Review

Recommender systems have evolved significantly in recent years. Traditional models were primarily collaborative, relying on user-item interactions. However, content-based systems gained prominence for scenarios with limited user data. These systems analyze textual or multimedia features of items to compute relevance. With the advent of NLP and machine learning, new methods such as TF-IDF and deep semantic embeddings have improved the

performance of content-based filtering. In this project, TF-IDF and cosine similarity were chosen for their efficiency, interpretability, and proven effectiveness in small to medium-scale datasets.

5. Methodology

The project follows a structured methodology involving data collection, preprocessing, feature extraction, similarity computation, sentiment analysis, and recommendation generation. Each step contributes to the overall robustness of the system.

The dataset used in this project contains news article titles and corresponding content. Text preprocessing involves lowercasing, punctuation removal, and stopword filtering. After cleaning, TF-IDF vectorization converts the textual data into numerical vectors, capturing the importance of each term in the corpus. Cosine similarity is then used to compute how closely articles are related based on their vector representations. Sentiment analysis is applied using TextBlob to assign a polarity score to each article. Finally, recommendations are generated based on both similarity scores and positive sentiment filtering.

6. System Architecture

The system architecture consists of several layers: data input, text preprocessing, feature extraction, similarity computation, sentiment analysis, and recommendation output. The process begins with raw text data, which is cleaned and transformed into a TF-IDF matrix. The cosine similarity function measures the closeness of each article to others. Sentiment analysis further refines recommendations by focusing on positively rated articles.

7. Implementation

The project is implemented using Python due to its extensive library support for machine learning and NLP. Key libraries include pandas for data manipulation, scikit-learn for TF-IDF vectorization and similarity calculations, and TextBlob for sentiment analysis. The implementation follows a modular approach, with separate functions handling preprocessing, similarity computation, keyword extraction, and sentiment evaluation.

8. Algorithms Used

The system utilizes two major algorithms: Term Frequency–Inverse Document Frequency (TF-IDF) for feature extraction and cosine similarity for measuring text similarity. TF-IDF assigns weights to words based on their frequency within an article and their rarity across the dataset. Cosine similarity, in turn, computes the cosine of the angle between two TF-IDF vectors, yielding a value between 0 and 1 that represents similarity. Additionally, sentiment polarity is calculated using TextBlob, which leverages a lexical approach to determine the emotional tone of a text.

9. Results and Discussion

The similarity matrix generated by the model highlights how closely related each article is to others. Articles sharing common topics, such as artificial intelligence or economics, yield higher similarity scores. The sentiment analysis adds another layer of interpretation by classifying articles as positive, negative, or neutral. The system's recommendation output successfully combines these elements, providing users with a list of similar and positively toned articles. For example, an article about 'AI transforming healthcare industry' yields related recommendations such as 'AI used to detect diseases earlier' and 'New breakthrough in cancer research.'

10. Evaluation Metrics

The model's performance can be evaluated using standard recommender system metrics such as precision, recall, and F1-score. Since this project focuses on content similarity, qualitative evaluation through expert assessment and user satisfaction feedback is also essential. The sentiment filter improves user satisfaction by aligning recommendations with emotional preferences.

11. Challenges Faced

One of the main challenges encountered during development was the limited size and diversity of the dataset. Small datasets restrict the generalization capability of the model. Additionally, sentiment analysis can sometimes misclassify neutral or context-dependent statements due to the simplistic nature of lexical polarity models. Another issue involved balancing TF-IDF feature extraction for short titles and longer article bodies. Despite these limitations, the system performed reliably within the test environment.

12. Future Enhancements

Future work could explore advanced deep learning approaches such as BERT or GPT-based embeddings to capture richer semantic relationships. Incorporating user profiling, personalization through interaction data, and real-time recommendation updates could further enhance system performance. Integrating multilingual support and topic modeling would also make the system more versatile and globally applicable.

13. Conclusion

The Advanced News Recommender System demonstrates the potential of combining NLP, sentiment analysis, and machine learning techniques for intelligent content recommendation. Through effective preprocessing, TF-IDF representation, and cosine similarity measurement, the system can successfully identify related articles. Sentiment-based filtering further enhances relevance and user engagement. Overall, the project

achieves its goal of building a robust and explainable news recommendation framework suitable for expansion and real-world application.

14. References

1. Pedregosa et al., 'Scikit-learn: Machine Learning in Python,' Journal of Machine Learning Research, 2011.
2. Steven Bird, Edward Loper, and Ewan Klein, 'Natural Language Processing with Python,' O'Reilly Media, 2009.
3. TextBlob Documentation: <https://textblob.readthedocs.io>
4. Scikit-learn Documentation: <https://scikit-learn.org/>
5. NLTK Documentation: <https://www.nltk.org/>

15. Appendix

This section summarizes the experimental setup and configurations used. The project was executed in a Python environment using Google Colab. All preprocessing and computation steps were performed using open-source libraries. The dataset consisted of six sample news articles focusing on topics such as artificial intelligence, healthcare, economics, and technology.